

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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PIPESIM Process Steady State Simulation Report

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R01	15-06-22	Issued for Review
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Revision Philosophy:

- All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-011

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

References

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-001	Process Flow Scheme (PFS)
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 (sheet 1 to 8)	Process Engineering Flow Scheme (PEFS)
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipe Wall Thickness Calculation Report
	FISAS ESIA Draft Report (November 2020)

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Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials
BARG	Bar Gauge
BOD	Basis of Design
BPD	Barrel Per Day
BCPD	Barrel of Condensate Per Day
BWPD	Barrel of Water Per Day
EPCI	Engineering, Procurement, Construction, Installation
FEED	Front End Engineering Design
MMSCFD	Million Standard Cubic Feet Per Day
NPS	Nominal Pipe Size
OD	Outside Diameter
WT	Wall Thickness

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in Figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

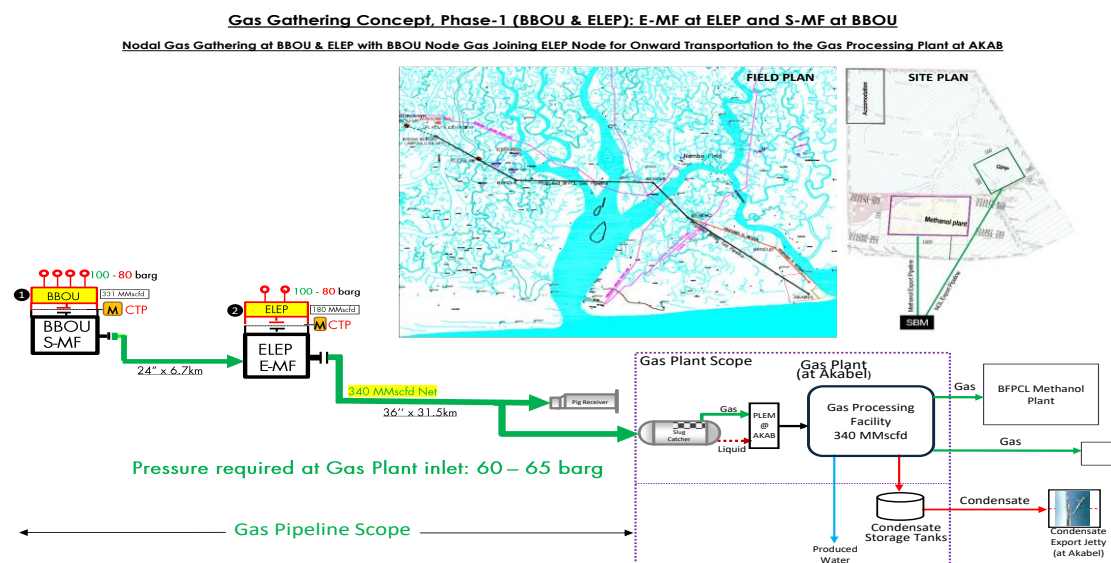


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to validate the proposed pipeline sizes during the conceptual engineering based on the available pipeline route survey map. To further carry out sensitivity analysis on other pipeline sizes for the entire Gas Gathering Pipeline Systems starting from the early gas production to the future 1 BCF gas production.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
API RECOMMENDED PRACTICE 14E	Recommended Practice for Design and Installation of Offshore Production Platform Piping Systems
ASME B31.8	Gas Transmission and Distribution Piping Systems
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Design Units of Measurement

Variables	Units
Barg	Bar Gauge (Pressure)
Bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
BCF	Billion Cubic Feet
Bopd	Barrels of Oil per day
Bpd or B/D	Barrel per day
BS&W	Base Sediment and Water
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade

Variables	Units
km	kilometre
m	meter
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousand Barrels of per Day
Mbwpd	Thousand Barrels of Water per Day
MJ	MegaJoule
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Part Per Million
psig	Pounds Per Square Inch Gauge (Pressure)
Component	Vol%, mol%, wt ppm
Liquid flow	Kg/h, m ³ /h
Steam flow	Kg/h
Level	%, m, mm
Density	Kg/m ³
Conductivity	μs/cm
Weight	Kg, t
Distance (length)	m, mm
Power	W, KW
Force	N, kN
Current	A, mA
Energy	MJ, kJ
Sound	dBA
Frequency	HZ

3. Design Basis

This section summarizes the assumptions, methodology and basis to obtain the various results during analysis of the Gas Gathering System. Based upon this analysis, a recommended pipeline systems will be considered for FEED phase of design.

3.1. Production Profile

The pipeline system has been designed to achieve the production forecast detailed below. The production forecast includes the production from the OPUG, BBOU, SEIB, NUNR and ELEPA fields respectively, taken from the field manifolds after commingling the individual production wells.

Different production years were considered for each of the NAG producing field and considerations were given to the max. gas, max. condensate and max. water production for different scenarios of the simulation, as summarized in Fig 2.1 below.

The available manifold pressure as per the GSPA is between 80 barg to 100 barg.

The actual required flowing tubing head pressure (FTHP) at manifold outlet will be determine during pipeline sizing by meeting the delivery pressure range of 60 barg to 65 barg as explained in section 3.4.

The production profile as given in the Tables 3.1 through 3.3 below were used for the hydraulic analysis.

Table 3.1: Production Profile Summary

BBOU Production			
Year	Gas (MMSCFD)	Condensate (BCPD)	Water (BWPD)
7	305	1217	9002
9	331	1323	8038
18	174	16226	1667
ELEPA Production			
14	386	22,905	16,513
18	382	34,433	5,014
OPUG Production			
16	154	9,082	3,859
18	174	16,226	1,667
20	182	15,606	2,775
SEIB Production			

BBOU Production			
Year	Gas (MMSCFD)	Condensate (BCPD)	Water (BWPD)
16	154	9,082	3,859
20	182	15,606	2,775
NUNR Production			
13	245	14,583	10,844
OTHERS Production			
	1,000	55000	3000

The detailed production forecast is also shown in Table 3.2 and 3.3 below.

Table 3.2: Detailed Production Forecast of BBOU Node (BBOU and OPUG)

BBOU TO ELEPA PIPELINE SEGMENT (24" x 6.4km)									
Year	BBOU Node			BUBOUWE BOU			OPUGBENE		
	TOTAL FLUID RATE								
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
0	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	181	944	907	181	944	907	0	0	0
2	181	878	907	181	878	907	0	0	0
3	181	788	907	181	788	907	0	0	0
4	181	742	7,855	181	742	7,855	0	0	0
5	181	707	7,547	181	707	7,547	0	0	0
6	215	840	8,491	215	840	8,491	0	0	0
7	305	1,217	9,002	305	1,217	9,002	0	0	0
8	327	1,323	8,193	327	1,323	8,193	0	0	0
9	331	1,338	8,038	331	1,338	8,038	0	0	0
10	93	1,763	2,527	67	239	2,397	26	1,525	129
11	75	3,242	1,301	21	59	1,031	54	3,183	270
12	74	3,457	1,074	16	42	784	58	3,416	290
13	71	4,084	1,513	2	4	85	69	4,080	1,429
14	71	4,161	3,048	0	0	0	71	4,161	3,048
15	94	5,541	3,748	0	0	0	94	5,541	3,748
16	154	9,082	3,859	0	0	0	154	9,082	3,859
17	170	12,115	3,148	0	0	0	170	12,115	3,148
18	174	16,226	1,667	0	0	0	174	16,226	1,667
19	177	15,812	1,558	0	0	0	177	15,812	1,558
20	182	15,606	2,775	0	0	0	182	15,606	2,775

ELEPA TO AKABEL PIPELINE SEGMENT (36" x 31.5km)																			
ELEP Node																			
year	TOTAL FLUID RATE			BUBOUWE BOU			ELEPA			NUN RIVER			SEIB			OPUG			
	MMscf/d	bcpd	bwpd	MMscf/d	bcpd	bwpd	MMscf/d	bcpd	bwpd	MMscf/d	bcpd	bwpd	MMscf/d	bcpd	bwpd	MMscf/d	bcpd	bwpd	
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	
1	361	11,032	1,807	181	944	907	180	10,088	900	0	0	0	0	0	0	0	0	0	
2	361	10,965	1,807	181	878	907	180	10,088	900	0	0	0	0	0	0	0	0	0	
3	361	10,876	1,807	181	788	907	180	10,088	900	0	0	0	0	0	0	0	0	0	
4	361	10,830	15,835	181	742	7,855	180	10,088	7,980	0	0	0	0	0	0	0	0	0	
5	361	10,794	15,213	181	707	7,547	180	10,088	7,666	0	0	0	0	0	0	0	0	0	
6	360	8,985	13,393	215	840	8,491	145	8,146	4,902	0	0	0	0	0	0	0	0	0	
7	357	4,128	11,599	305	1,217	9,002	52	2,911	2,597	0	0	0	0	0	0	0	0	0	
8	356	2,953	9,647	327	1,323	8,193	29	1,630	1,454	0	0	0	0	0	0	0	0	0	
9	356	2,724	9,275	331	1,338	8,038	25	1,386	1,236	0	0	0	0	0	0	0	0	0	
10	382	18,828	4,837	67	239	2,397	19	1,076	960	244	14,464	1,221	26	1,525	129	26	1,525	129	
11	385	21,637	3,357	21	59	1,031	11	628	561	245	14,583	1,225	54	3,183	270	54	3,183	270	
12	385	21,933	3,015	16	42	784	9	478	426	245	14,583	1,225	58	3,416	290	58	3,416	290	
13	386	22,799	13,832	2	4	85	1	52	46	245	14,583	10,844	69	4,080	1,429	69	4,080	1,429	
14	386	22,905	16,513	0	0	0	0	0	0	245	14,583	10,417	71	4,161	3,048	71	4,161	3,048	

15		384	21,99 6	14,14 9		0	0	0		0	0	0		196	10,91 4	6,653		94	5,541	3,748		94	5,541	3,748
16		379	22,32 3	11,24 6		0	0	0		0	0	0		71	4,159	3,528		154	9,082	3,859		154	9,082	3,859
17		380	26,55 8	8,272		0	0	0		0	0	0		40	2,329	1,976		170	12,11 5	3,148		170	12,11 5	3,148
18		382	34,43 3	5,014		0	0	0		0	0	0		34	1,980	1,680		174	16,22 6	1,667		174	16,22 6	1,667
19		380	33,16 1	4,421		0	0	0		0	0	0		26	1,538	1,304		177	15,81 2	1,558		177	15,81 2	1,558
20		379	32,11 1	6,313		0	0	0		0	0	0		15	898	762		182	15,60 6	2,775		182	15,60 6	2,775

Table 3.3: Detailed Production Forecast of ELEPA Node

3.2. Fluid Composition

The fluid composition used for the simulation for the different field are below in fig 3.4.

Table 3.4: Fluid Composition

NODE	ELEPA TO AKABEL PIPELINE SEGMENT Fluid Composition (%)						BBOU TO ELEPA PIPELINE SEGMENT Fluid Composition (%)		
Components	BBOU	ELEP	NUNR	SEIB	OPUG	P/L Segment	BBOU	OPUG	P/L Segment
N ₂	0.07	NA	NA	NA	0.07	0.03514	0.07	0.07	0.0700
CO ₂	0.59	2.18	0.77	0.77	0.59	1.38185	0.59	0.59	0.5900
C ₁	92.92	84.165	72.83	72.83	92.92	88.5598	92.92	92.92	92.9200
C ₂	3.16	6.17	7.68	7.68	3.16	4.65904	3.16	3.16	3.1600
C ₃	1.01	2.32	4.85	4.85	1.01	1.66241	1.01	1.01	1.0100
iC ₄	0.61	0.635	1.46	1.46	0.61	0.62245	0.61	0.61	0.6100
nC ₄	0.44	0.65	1.97	1.97	0.44	0.54458	0.44	0.44	0.4400
iC ₅	0.25	0.35	0.95	0.95	0.25	0.2998	0.25	0.25	0.2500
nC ₅	0.19	0.245	0.76	0.76	0.19	0.21739	0.19	0.19	0.1900
C ₆	0.23	0.465	1.03	1.03	0.23	0.34703	0.23	0.23	0.2300
C ₇	0.19	0.635	1.41	1.41	0.19	0.41162	0.19	0.19	0.1900
C ₈	0.04	0.59	1.61	1.61	0.04	0.31391	0.04	0.04	0.0400
C ₉	0.03	0.24	0.9	0.9	0.03	0.13458	0.03	0.03	0.0300
C ₁₀	0.02	0.185	0.65	0.65	0.02	0.10217	0.02	0.02	0.0200
C ₁₁	0.03	0.14	0.47	0.47	0.03	0.08478	0.03	0.03	0.0300
C ₁₂₊	0.068	0.135	0.37	0.37	0.068	0.10137	0.07	0.07	0.0680
C ₁₃	0.047	0.14	0.34	0.34	0.047	0.09332	0.05	0.05	0.0470
C ₁₄	0.032	0.13	0.3	0.3	0.032	0.08081	0.03	0.03	0.0320
C ₁₅	0.022	0.115	0.26	0.26	0.022	0.06832	0.02	0.02	0.0220
C ₁₆	0.015	0.09	0.21	0.21	0.015	0.05235	0.02	0.02	0.0150
C ₁₇	0.01	0.07	0.16	0.16	0.01	0.03988	0.01	0.01	0.0100
C ₁₈	0.007	0.065	0.14	0.14	0.007	0.03589	0.01	0.01	0.0070
C ₁₉	0.016	0.04	0.08	0.08	0.016	0.02795	0.02	0.02	0.0160
C ₂₀₊	NA	0.25	0.8	0.8	NA	0.12451	NA	NA	NA
Total	100.00	100.01	100.00	100.00	100.00	100.00	100.00	100.00	100.00

3.3. Ambient Conditions

The following were considered as the ambient condition:

Table 3.5: Ambient Conditions

	Units	
Minimum ambient temperature	°C	18
Minimum Soil temperature	°C	23
Minimum swamp temperature	°C	23

Manifold Conditions

The manifold conditions for the various fields are presented in Table 3.6 below. This is based on current available data. The lowest manifold temperature of 45°C is used for the manifold temperature as the inlet to the pipeline system.

Table 3.6: Manifold Conditions

S/N	Field	Manifold Flowing Temperature (°C)	Manifold Flowing Pressure (barg)
1	BBOU	45 – 95	80 – 100
2	ELEPA	45 – 95	80 – 100
3	OPUG	45 – 95	80 – 100
4	SEIB	45 – 95	80 – 100
5	NUNR	45 – 95	80 – 100

3.4. Pipeline Delivery Pressure

The gas from different fields will delivered to the Gas Plant at Akabel. The pipeline system gas delivery pressure is between 60 barg to 65 barg. The delivery pressure is the inlet pressure requirement at Akabel Gas Plant.

3.5. Pipeline Profile

The pipeline topography is as shown in the graph Fig 3.1 and Fig 3.2 below. The pipeline burial depth is 1.5 meter below the msl. The wellhead platform for the producing fields is assumed to be elevated (5m above the msl).

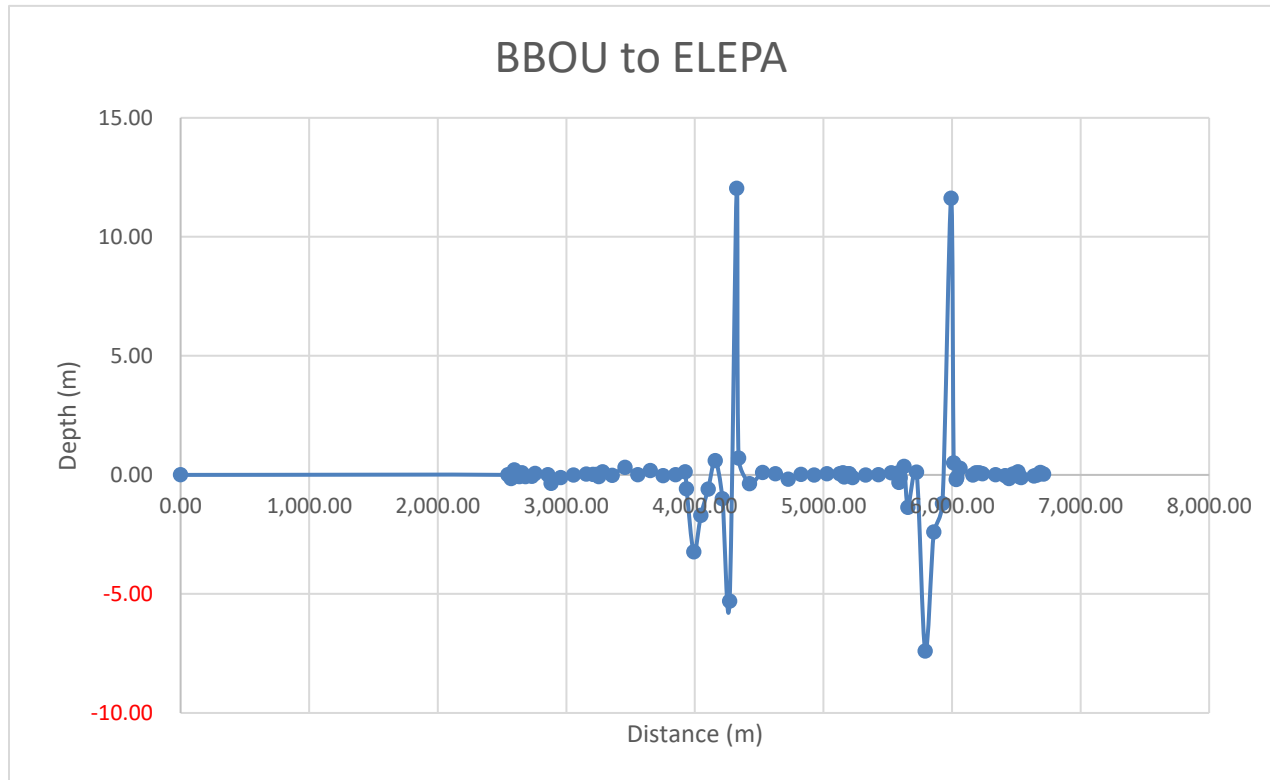


Fig 3.1: Pipeline profile from BBOU to ELEPA

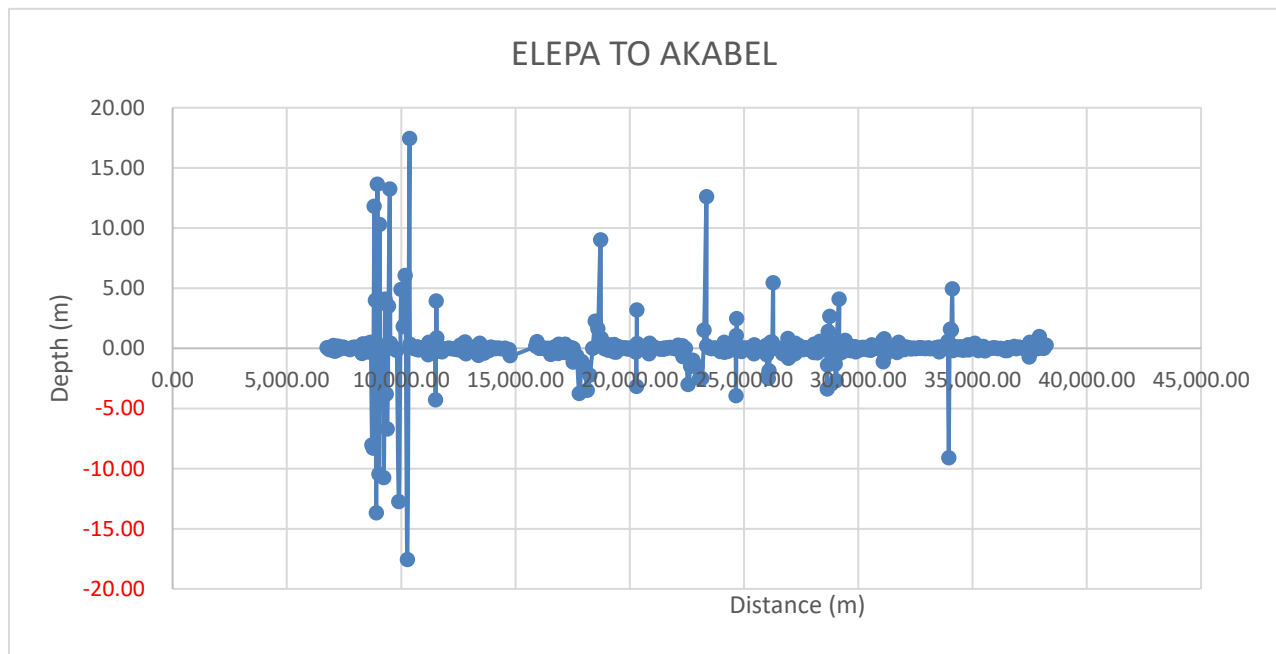


Fig 3.2: Pipeline profile from ELEPA to AKABEL

3.6. Pipeline Characteristics

The pipeline wall thickness used for the analysis are presented below in the table below:

Table 3.7: pipeline characteristics

Wall Section	Material	Outer Diameter mm (in)	Thickness mm (in)	Inside Diameter mm (in)
Pipeline	24 - inch Carbon Steel	610 (24)	15.88 (0.625197)	577.85 (22.75)
	28 - inch Carbon Steel	711 (28)	17.48 (0.688189)	676.15 (26.62)
	36 - inch Carbon Steel	914 (36)	23.83 (0.938189)	866.65 (34.12)

A roughness of 0.0018in is used for the analysis.

3.7. Pipeline Sizing Criteria

The following aspects has been considered while sizing the pipeline from OPUG to BBOU, BBOU to ELEPA, ELEPA to AKABEL and from SEIB to NUNR to ELEPA to AKABEL.

1. System deliverability this includes delivering a minimum of 60 barg at AKABEL with a minimum inlet pressure 80 barg at field manifold. This is applicable to all scenarios of the pipeline network to AKABEL.
2. Deliverability of the maximum condensate and water volume alongside the gas volume to AKABEL maintaining the desired delivery pressure of 60 barg minimum.
3. Erosional Velocity Ratio (EVR) of < 1 , to avoid the erosion of the pipeline. EVR is a ratio of fluid velocity to the erosional velocity calculated from the erosion model, API 14E.

4. Approach To Work

This section describes the process simulation structure and methodology. The process description is however based on the gas production scenarios coming differently to supply gas to AKABEL as developed during the conceptual stage of this project and aligning with the pipeline route survey map/report.

4.1. Simulation Tools

Version 2017.1 of PIPESIM Hydraulics software was used in performing the hydraulics simulation of the pipelines systems. UNISIM R460.1 was used to create the gas, condensate and water compositions from the available bulk fluid compositions.

4.2. Steady State Simulations

For the hydraulic analysis simulations, the correlation for both vertical and horizontal Pipes was selected as Hagedorn & Brown and Beggs & Bill Revised respectively. The Table 4.1 below shows the thermodynamics simulation set up data for the Pipesim simulation set up.

Table 4.1 – Pipesim Simulation Set-up

Simulation Set up	Method
Simulation Type	Network Model
Fluid Type	Compositional
Property Package (Equation of State-EOS)	Peng Robinson

3-Phase Composition: Since the available composition was the bulk fluid composition from the manifold and the available production profile/flowrate was the gas, condensate and water flowrate, UNISIM R460.1 simulation tool was used to create the gas, condensate and water compositions for proper data input and analysis

The bulk composition was saturated with water and passed through a three-phase separator operating at a 100 barg pressure for both BBOU Node and ELEPA Node, Fig 4.1 and 4.2.

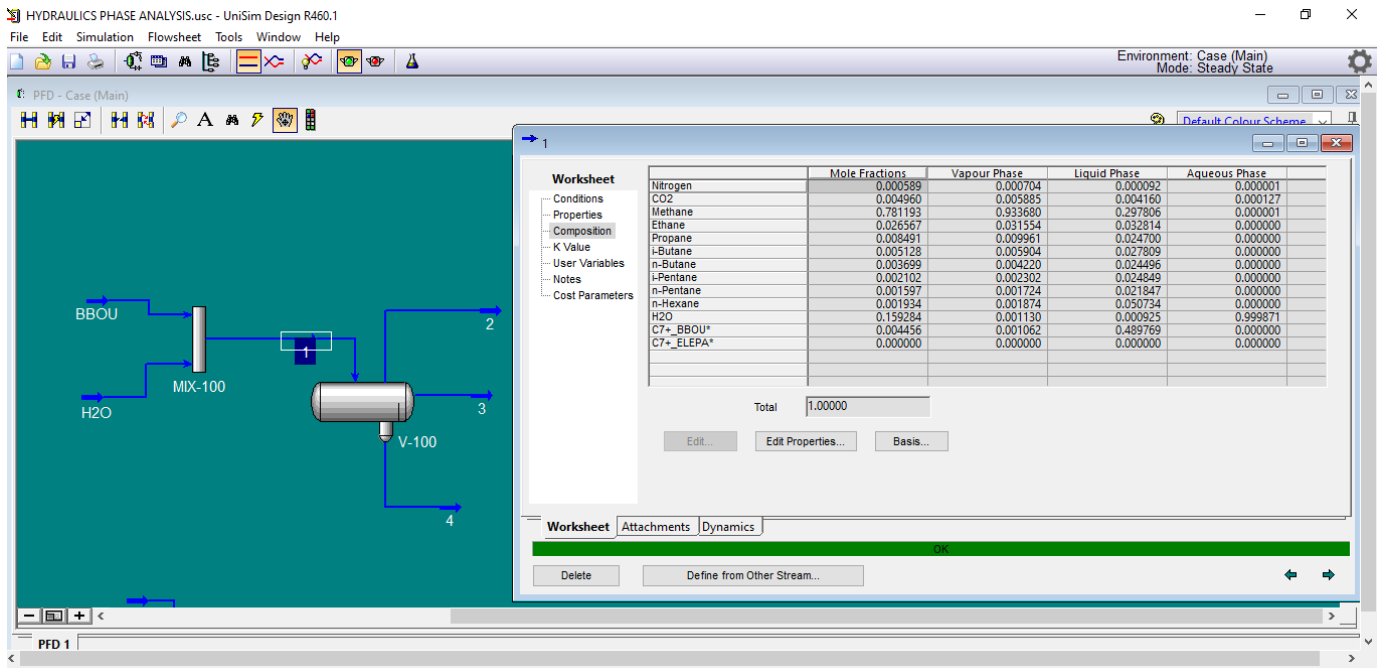


Fig 4.1: BBOU Node (BBOU/OPUG) 3-Phase Composition

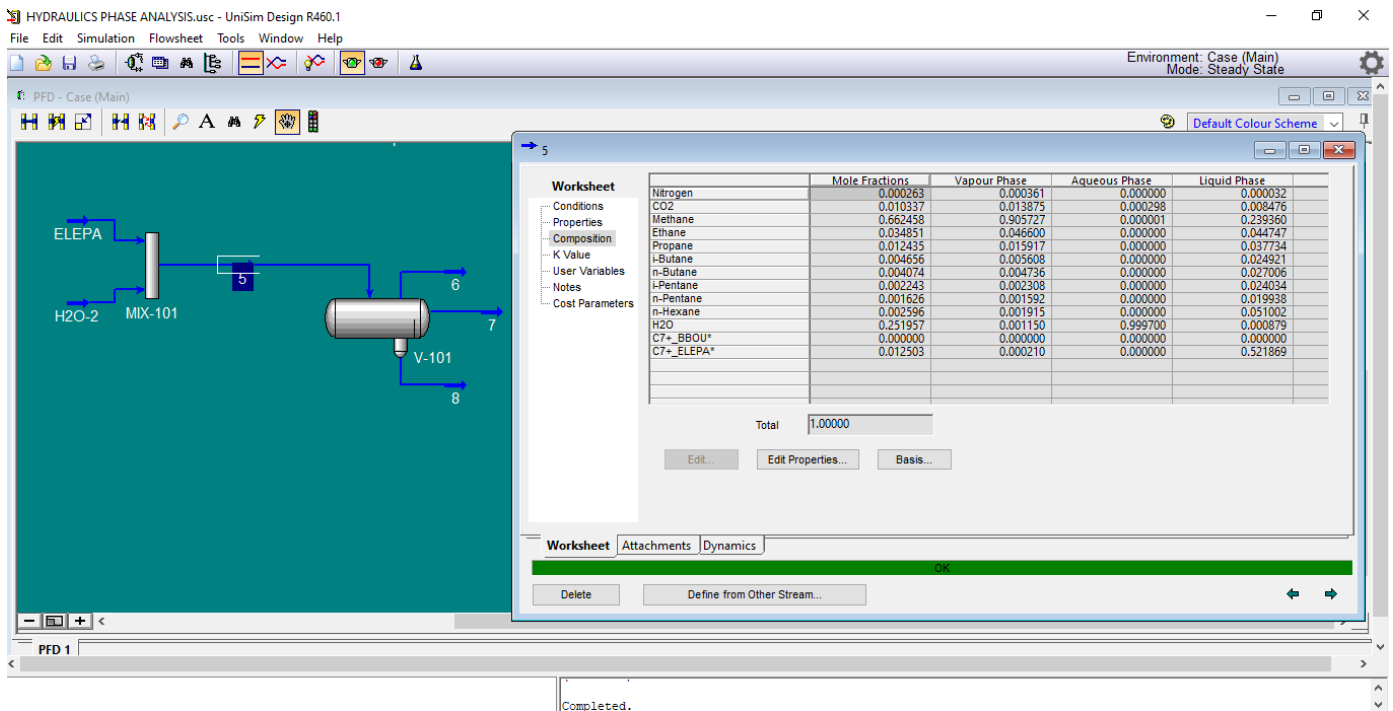


Fig 4.2: ELEPA Node (ELEPA/SEIB/NUNR) 3-Phase Composition

4.3. Simulation Scenarios

Different productions were considered from different fields, as shown in the Gas Gathering Pipeline Schematics fig 4.3. Each of the production scenarios from either of the fields; OPUG, BBOU, SEIB, NUNR and ELEPA (or a combination of two or more) were considered at different Flowing Tubing Head Pressure (FTHP) ranging from 100 barg to 80 barg in order to deliver at a

pressure range of 60barg to 65barg to Akabel. At a minimum FTHP of 80 barg from the field manifolds, different pipeline sizes were considered to deliver the bulk flow to Akabel at a minimum pressure of 60 barg

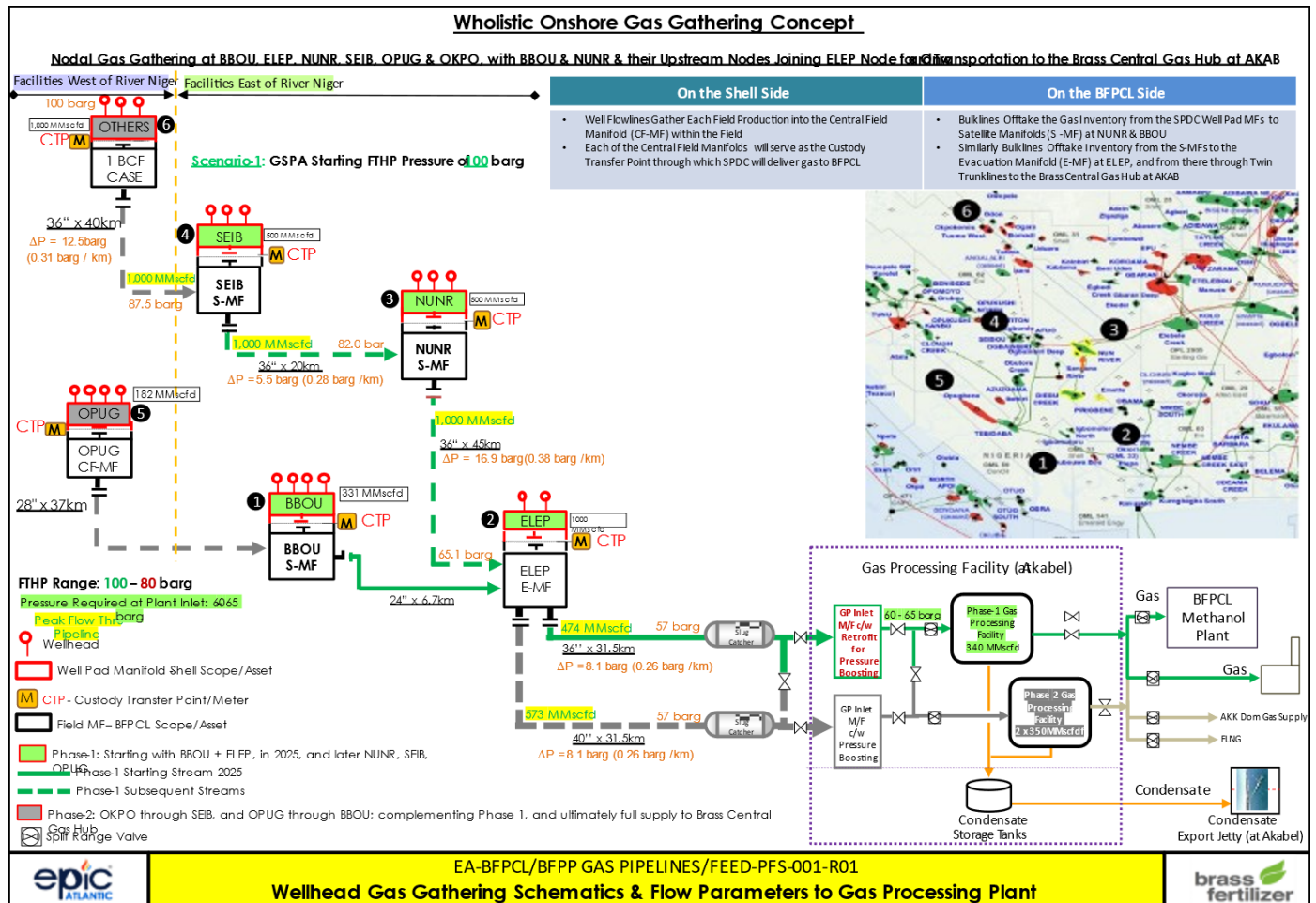


Fig 4.3: Gas Gathering Pipeline Schematics

4.3.1. BBOU to ELEPA to AKABEL Scenario

This involves the production from the BBOU to AKABEL through ELEPA. The production considered maximum rate of 331 MMSCFD, 16,200 BCPD and 9,000 BWPD from the BBOU and different Flowing Tubing Head Pressure (FTHP) ranging from 100 barg to 80 barg. This production form the basis for hydraulic analysis of the pipeline from BBOU to ELEPA, which is 6.7km apart. The simulation snapshot is shown in Fig 4.4.

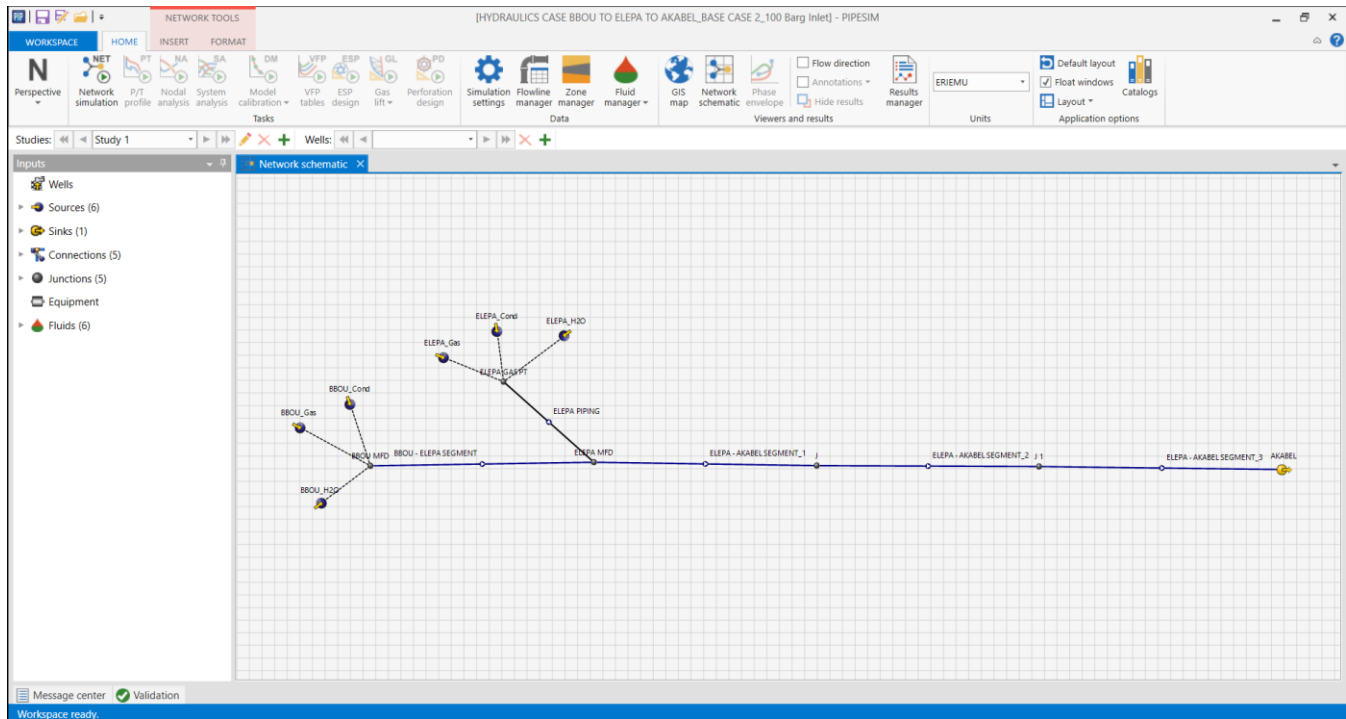


Fig 4.4: BBOU to ELEPA to AKABEL Hydraulics Simulation Snapshot.

4.3.2. ELEPA to AKABEL Scenario

Here, the production from the ELEPA to AKABEL at a distance of 31.5km were considered. The production considered at ELEPA were a combination of productions from different fields as per the production year and combined at ELEPA. The combined flow is expected to have enough pressure to arrive at AKABEL at a minimum pressure of 60 barg. A combined production rate of 386 MMSCFD, 34.4k BCPD and 9k BWPD was expected to flow from the ELEPA to AKABEL.

4.3.3. OPUG to BBOU to ELEPA to AKABEL Scenario

This was a scenario where there was production from OPUG to AKABEL via BBOU and ELEPA. The peak production rate of 182 MMSCFD, 16.2k BCPD and 3.9k BWPD from OPUG as shown in Fig 4.5 was used for the hydraulic analysis of the 37km pipeline from OPUG to BBOU.

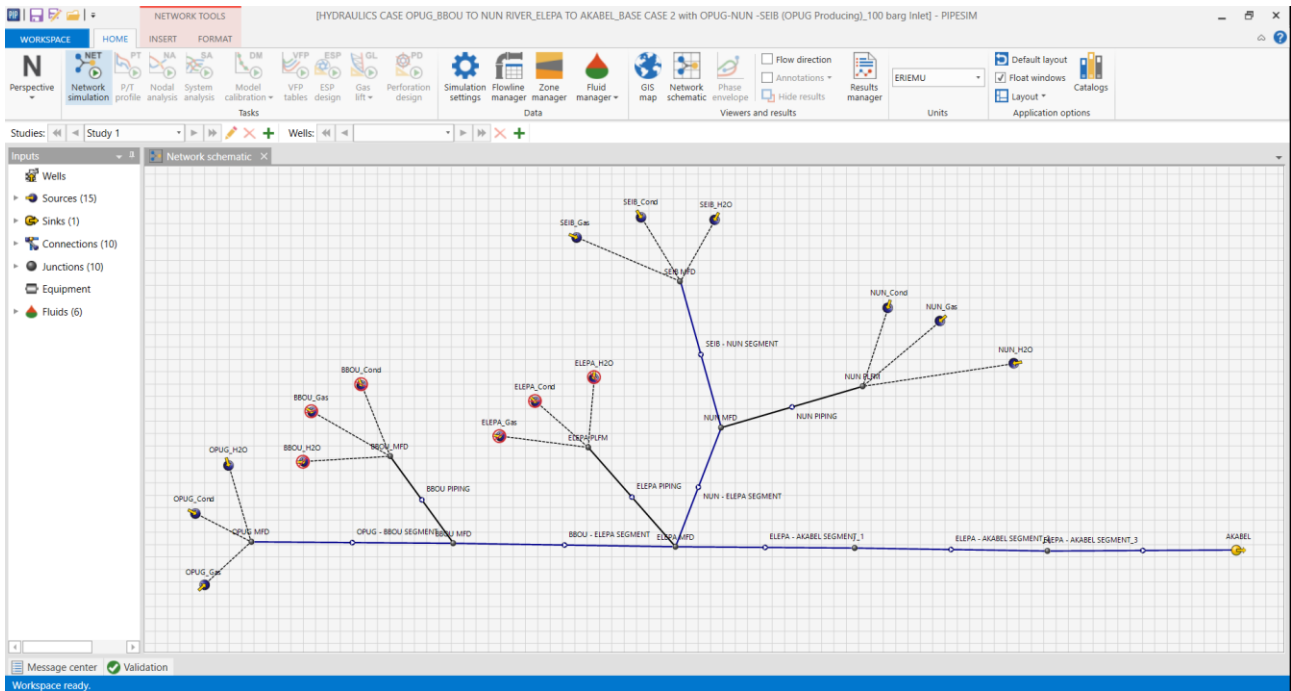


Fig 4.5: OPUG to BBOU to ELEPA to AKABEL Hydraulics Simulation Snapshot.

4.3.4. SEIB to NUNR to ELEPA to AKABEL Scenario

This production scenarios covers the SEIB peak production of 182 MMSCFD, 16.2k BCPD and 3.9k BWPD and NUNR peak production of 245 MMSCFD, 14.6k BCPD and 10.8k BWPD. The field is 20km to the NUNR and the NUNR is 45km to ELEPA.

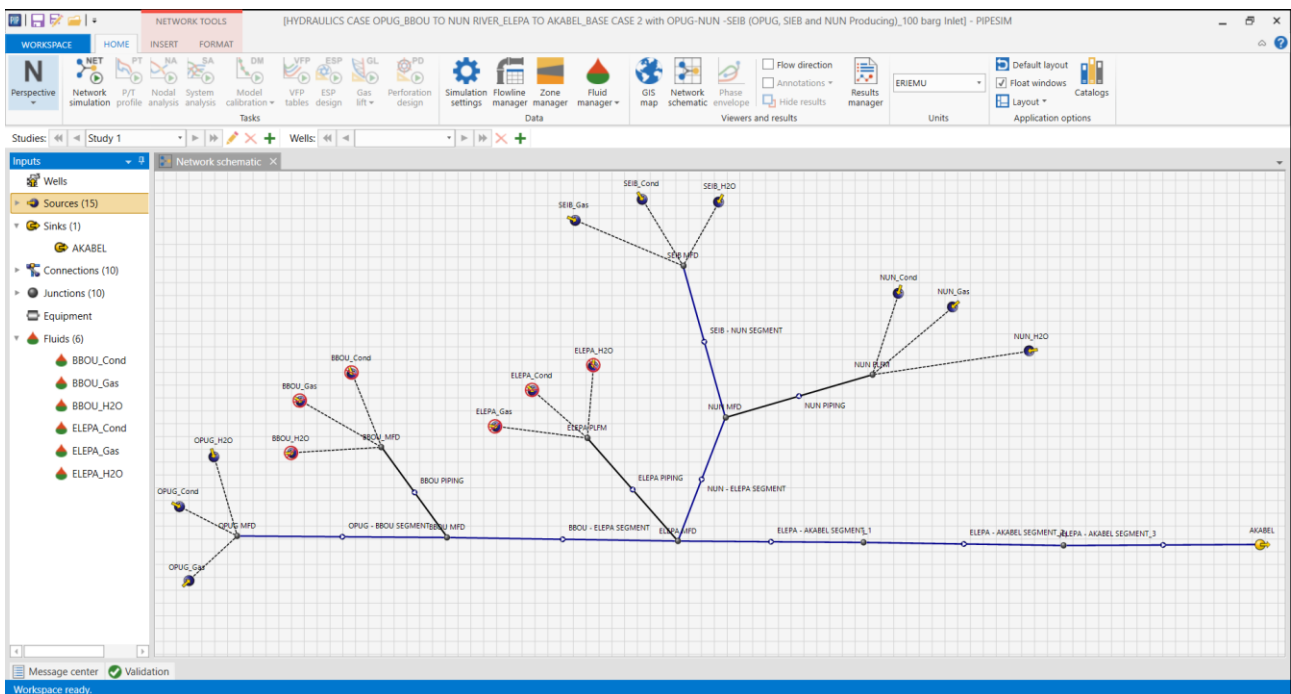


Fig 4.6: SEIB to NUNR to ELEPA to AKABEL Hydraulics Simulation Snapshot.

4.3.5. OTHERS to SEIB to NUNR to ELEPA to AKABEL Scenario

Due to the requirement for the future production of 1,000 MMSCFD (1BCFPD) from Others/SEIB/NUNR fields, considerations were given to this production to AKABEL, with an inlet pressure range of 80 barg to 100 barg. The peak production rate of 1,000 MMSCFD, 55k BCPD and 3k BWPD from OTHERS field as shown in Fig 4.7. The 2-loop production systems from ELEPA to AKABEL were used to deliver the bulk flow from ELEPA to AKABEL because the 36" ELEPA to AKABEL pipeline would have been existing prior to the commencement of the 1 BCF production, thus there is need for a second line to ensure adequate delivery to AKABEL.

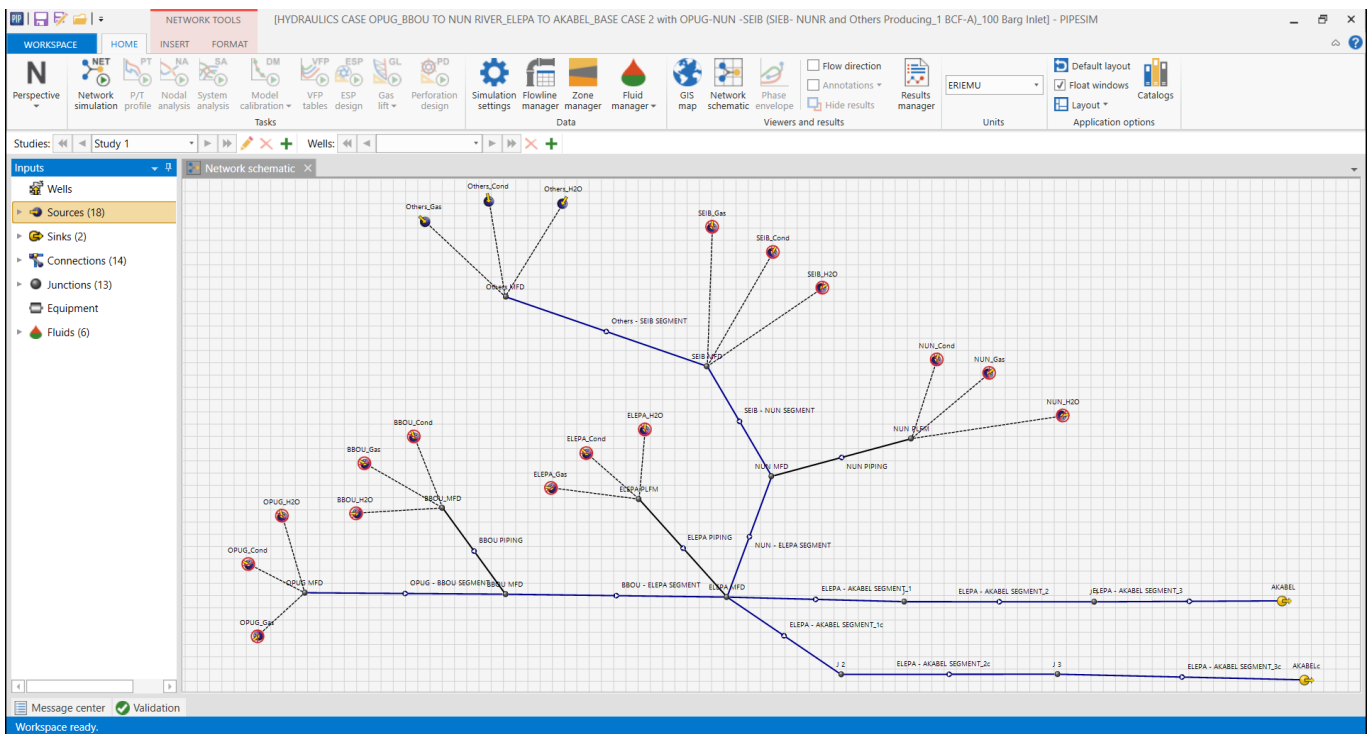


Fig 4.7: OTHERS to SEIB to NUNR to ELEPA to AKABEL Hydraulics Simulation Snapshot.

5. Pipesim Hydraulics Simulation Results

The result of the hydraulic analysis performed at the two different Manifold Inlet Flowing Tubing Head Pressure (FTHP) conditions of 100 barg and 80 barg and a minimum desired delivery pressure of 60 barg at AKABEL (except for 1 BCFD production only when the desired delivery pressure of 5 barg to 10 barg is required at AKABEL).

5.1. BBOU to ELEPA to AKABEL Pipeline Network System

Sensitivity analysis were performed at different AKABEL pressure scenario as shown in result Table 5.1 below.

BBOU -ELEPA-AKABEL Pipeline Hydraulic Analysis										
S/N	Simulation Case	Pipeline Size (in): BBOU TO ELEPA	Pipeline Size (in): ELEPA TO AKABEL	Inlet Press (barg): BBOU	Inlet Press (barg): ELEPA	Delivery Press (barg): AKABEL	Erotional Velocity Ratio (EVR<1, desired): BBOU to ELEPA	Erotional Velocity Ratio (EVR<1, desired): ELEPA to AKABEL	Velocity (ft/sec): BBOU to ELEPA	Velocity (ft/sec): ELEPA to AKABEL
1	1	24	28	80	77.12	67.00	0.393	0.305	17.25	10.48
2				100	98.62	90.00	0.348	0.261	13.48	10.35
3	2	24	28	80	78.19	62.92	0.125	0.440	4.05	18.30
4				100	97.75	84.66	0.112	0.378	3.24	13.39
5		24	36	100	97.75	87.92	0.112	0.226	3.22	7.85
4				80	78.19	68.02	0.125	0.259	4.02	10.28
Case 1: Production from BBOU Only										
Case 2: Production from BBOU and ELEPA (controlling case)										
Pipeline Distance: BBOU to ELEPA is 6.7km and ELEPA to AKABEL is 31.5km										

Fig 5.1: Sensitivity Analysis Result of BBOU – ELEPA – AKABEL Hydraulic Analysis.

From the above table, 24" pipeline size from the BBOU to ELEPA was selected. From ELEPA to AKABEL, 36" pipeline was selected as it will accommodate a larger from the SEIB-NUNR-ELEPA-AKABEL.

5.2. OPUG to BBOU to ELEPA to AKABEL Pipeline Network System

An integrated sensitivity analysis was performed at different pressure scenario as shown in result Table 5.2 below to ascertain the adequate pipeline size from OPUG to BBOU with an integrated effect of delivering a minimum of 60barg at AKABEL.

The hydraulics of OPUG to BBOU successfully favoured the 28" pipeline to adequately deliver greater than or equal to 60barg at Akabel via ELEPA as shown in result Table 5.2 below.

OPUG to BBOU MANIFOLD Pipeline Hydraulic Analysis														
S/N	Simulation Case	Pipeline Size (in): OPUG TO BBOU	Pipeline Size (in): BBOU TO ELEPA	Pipeline Size (in): ELEPA TO AKABEL	Inlet Press (barg): OPUG	Delivery Press (barg): BBOU MFD	Delivery Press (barg): ELEPA	Delivery Press (barg): AKABEL	Erosional Velocity Ratio (EVR<1, desired)			Velocity (ft/sec)		
									OPUG to BBOU	BBOU to ELEPA	ELEPA to AKABEL	OPUG to BBOU	BBOU to ELEPA	ELEPA to AKABEL
		24	24	28	100	95.17	93.55	65.57	0.216	0.219	0.398	7.58	7.74	15.78
					80	74.14	72.39	35	0.246	0.250	0.558	9.82	10.12	22.19
		24	24	32	100	95.17	93.55	75.09	0.217	0.218	0.298	7.60	7.72	11.10
					80	74.14	72.39	49.43	0.246	0.250	0.373	9.79	10.17	17.29
		24	24	36	100	95.17	93.55	80.13	0.218	0.219	0.220	7.70	7.79	7.97
					80	74.14	72.39	57.00	0.246	0.249	0.260	9.78	10.02	11.08
		28	24	36	100	98.14	96.54	83.50	0.156	0.215	0.215	5.42	7.48	7.60
					80	78.46	76.10	62.43	0.175	0.242	0.251	6.82	9.50	10.33
Pipeline Distance: OPUG to BBOU is 37km														

Fig 5.2: Sensitivity Analysis Result of OPUG – BBOU - ELEPA – AKABEL Hydraulic Analysis.

5.3. SEIB to NUNR to ELEPA to AKABEL Pipeline Network System

At a certain time in the life of the plant, production from the other fields would have been depleted and thus there is need for hydraulic analysis of pipeline producing from the SEIB to NUNR to ELEPA to AKABEL. The sensitivity analysis was performed at different pressure scenario as shown in result Table 5.3 below.

SEIB-NUNR to ELEPA MANIFOLD Pipeline Hydraulic Analysis														
S/N	Simulation Case	Pipeline Size (in): SEIB TO NUNR	Pipeline Size (in): NUNR TO ELEPA	Pipeline Size (in): ELEPA TO AKABEL	Inlet Press (barg): SEIB	Delivery Press (barg): NUNR	Delivery Press (barg) at ELEPA	Delivery Press (barg) at AKABEL	Erotrional Velocity Ratio (EVR<1, desired)			Velocity (ft/sec)		
									SEIB to NUNR	NUNR to ELEPA	ELEPA to AKABEL	SEIB to NUNR	NUNR to ELEPA	ELEPA to AKABEL
3	2	36	36	28	100	99.22	95.47	67.00	0.088	0.218	0.423	3.168	7.66	17.6
4					80	79.00	75.55	35.00	0.099	0.247	0.595	4.02	9.84	34.7
3		36	36	32	100	99.59	96.46	78.00	0.088	0.218	0.313	3.16	7.63	12.04
4					80	79.17	74.04	51.00	0.10	0.248	0.393	4.04	9.95	18.95
3		36	36	36	100	98.42	96.56	84.00	0.088	0.218	0.229	3.17	7.69	8.48
4					80	79.17	75.55	61.87	0.10	0.248	0.271	4.03	9.91	11.85
Case 2: SEIB and NUNR Production														
Pipeline Distance: SEIB to NUNR is 20km and NUNR to ELEPA is 45km														

Fig 5.3: Sensitivity Analysis Result of SEIB – NUNR - ELEPA – AKABEL Hydraulic Analysis.

With the production volumes coming from the SEIB to NUNR to ELEPA, it was confirmed that 36" pipeline from SEIB to NUNR and 36" pipeline from NUNR to ELEPA will be adequate for the such volume of gas and liquid production. A further hydraulics analysis confirmation was carried out on the ELEPA to AKABEL pipeline and the 36" pipeline was reaffirmed to be the required pipeline size.

5.4. OTHERS to SEIB to NUNR to ELEPA to AKABEL Pipeline Network System

There is need to increase the gas supply to 1 BCF. Sensitivity analysis was carried out on the pipeline system to confirm the suitable pipeline sizes that will deliver 1 BCF to Akabel at different pressure scenario as shown in result Table 5.4 below.

OTHERS-SEIB-NUNR to ELEPA MANIFOLD to AKABEL Pipeline Hydraulic Analysis (1 BCF)																						
S/N	Simulation Case	Pipeline Size (in): OTHERS to SEIB	Pipeline Size (in): SEIB to NUNR	Pipeline Size (in): NUNR TO ELEPA	Pipeline Size (in): ELEPA TO AKABEL (LOOP 1)	Pipeline Size (in): ELEPA TO AKABEL (LOOP 2)	Inlet Press (barg): OTHERS	Inlet Press (barg): SEIB	Inlet Press (barg): NUNR	Delivery Press (barg) at ELEPA MFD	Delivery Press (barg) at AKABEL (Loop 1_36")	Delivery Press (barg) at AKABEL (Loop 2_40")	Ertional Velocity Ratio (EVR<1, desired)					Velocity (ft/sec)				
													OTHERS to SEIB	SEIB to NUNR	NUNR to ELEPA	ELEPA TO AKABEL (LOOP 1)	ELEPA TO AKABEL (LOOP 2)	OTHERS to SEIB	SEIB to NUNR	NUNR to ELEPA	ELEPA TO AKABEL (LOOP 1)	ELEPA TO AKABEL (LOOP 2)
1	1 BCF Production	36	36	36	28	40	100	89.27	84.26	69.5	60.00	60.00	0.492	0.505	0.554	0.257	0.354	19.38	20.39	24.57	12.14	16.73
2							80	65.69	58.67	32.96	18.00	18.00	0.578	0.610	0.810	0.517	0.633	26.76	29.81	52.52	45.10	55.22
5		36	36	36	36	100	86.73	81.30	64.71	56.00	56.00	0.495	0.514	0.570	0.306	0.306	19.632	21.11	26.05	15	15.00	
6						80	65.25	57.96	30.55	17.00	17.00	0.580	0.614	0.842	0.560	0.560	26.95	30.2	56.72	50.28	50.28	
7		36	36	36	36	40	100	88.77	83.51	65.10	60.00	60.00	0.493	0.506	0.558	0.286	0.248	19.49	20.52	24.92	13.52	11.74
8							80	66.95	60.48	34.17	24.40	24.40	0.572	0.603	0.795	0.411	0.431	26.23	29.16	50.65	30.92	32.44
Pipeline Distance: OTHERS to SEIB is 40km																						

Pipeline Distance: OTHERS to SEIB is 40km

Fig 5.4: Sensitivity Analysis Result of OTHERS-SEIB-NUNR-ELEPA-AKABEL Hydraulic Analysis of 1 BCF.

The hydraulics considers an optimal arrival pressure that will require less power for compression at AKABEL with 2-loop production systems of already selected 36" pipeline and 40" from ELEPA to AKABEL as indicated in the table. The inlet pressures at the OTHER field are between 100barg to 80 barg.

6. Recommendations

The result of the hydraulics analysis of the Gas Gathering Pipeline Systems confirms and recommend the following pipeline sizes:

- BBOU to ELEPA 6.7km pipeline is recommended to be 24" since it satisfies the requirement of delivering the gas to Akabel at a pressure above 60 barg.
- 28" pipeline is recommended for OPUG to BBOU 37km
- For the gas transmission pipeline from ELEPA to AKABEL, 36" pipeline is recommended.
- 36" pipeline is recommended for gas transmission from OTHERS (1 BCF of gas) to SEIB to NUNNR to ELEPA. For accurate and appropriate delivery of the 1 BCF of gas to AKABEL facilities, a 40" pipeline is recommended to be looped with the earlier recommended 36" pipeline to the gas facilities to AKABEL.

The recommended pipeline schematics is shown in Figs 6.1 and 6.2.

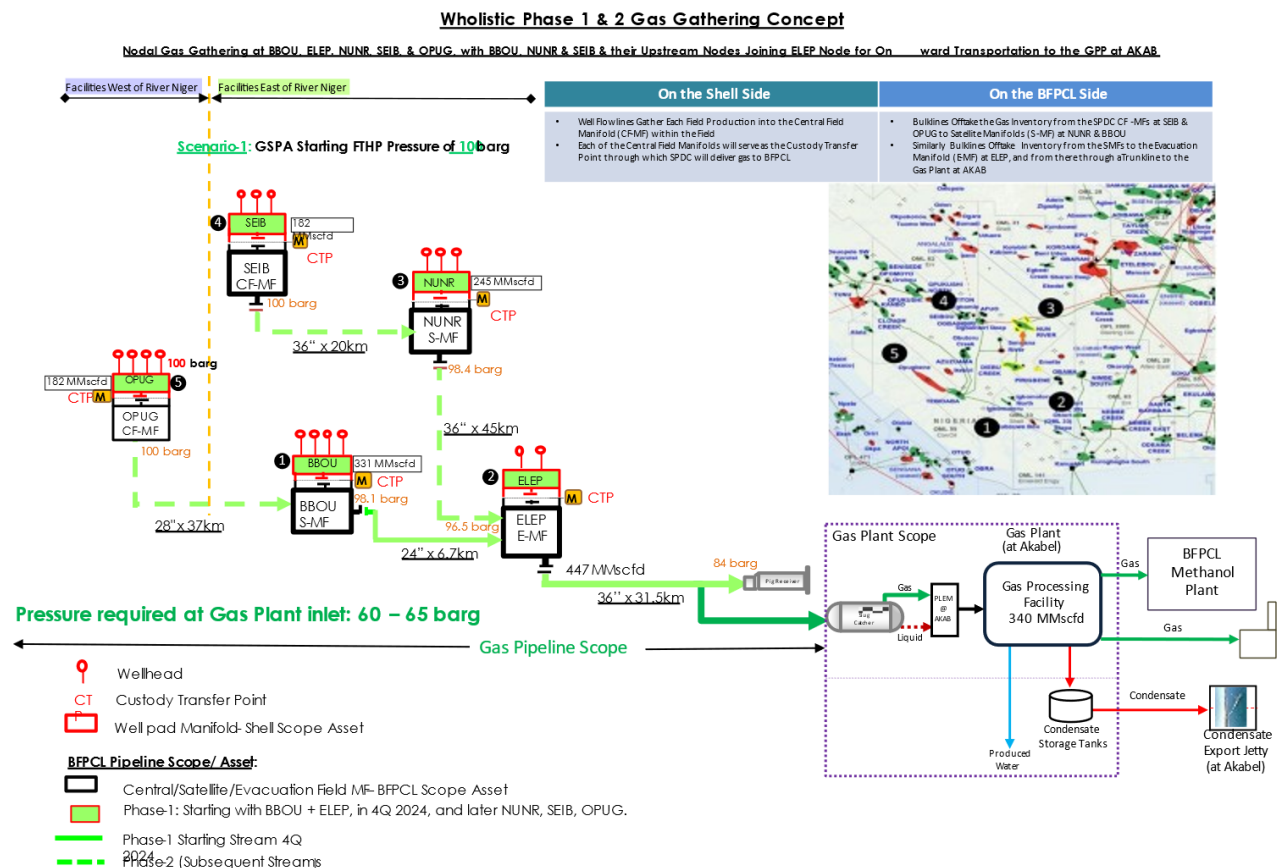


Fig 6.1: Recommended pipeline schematics

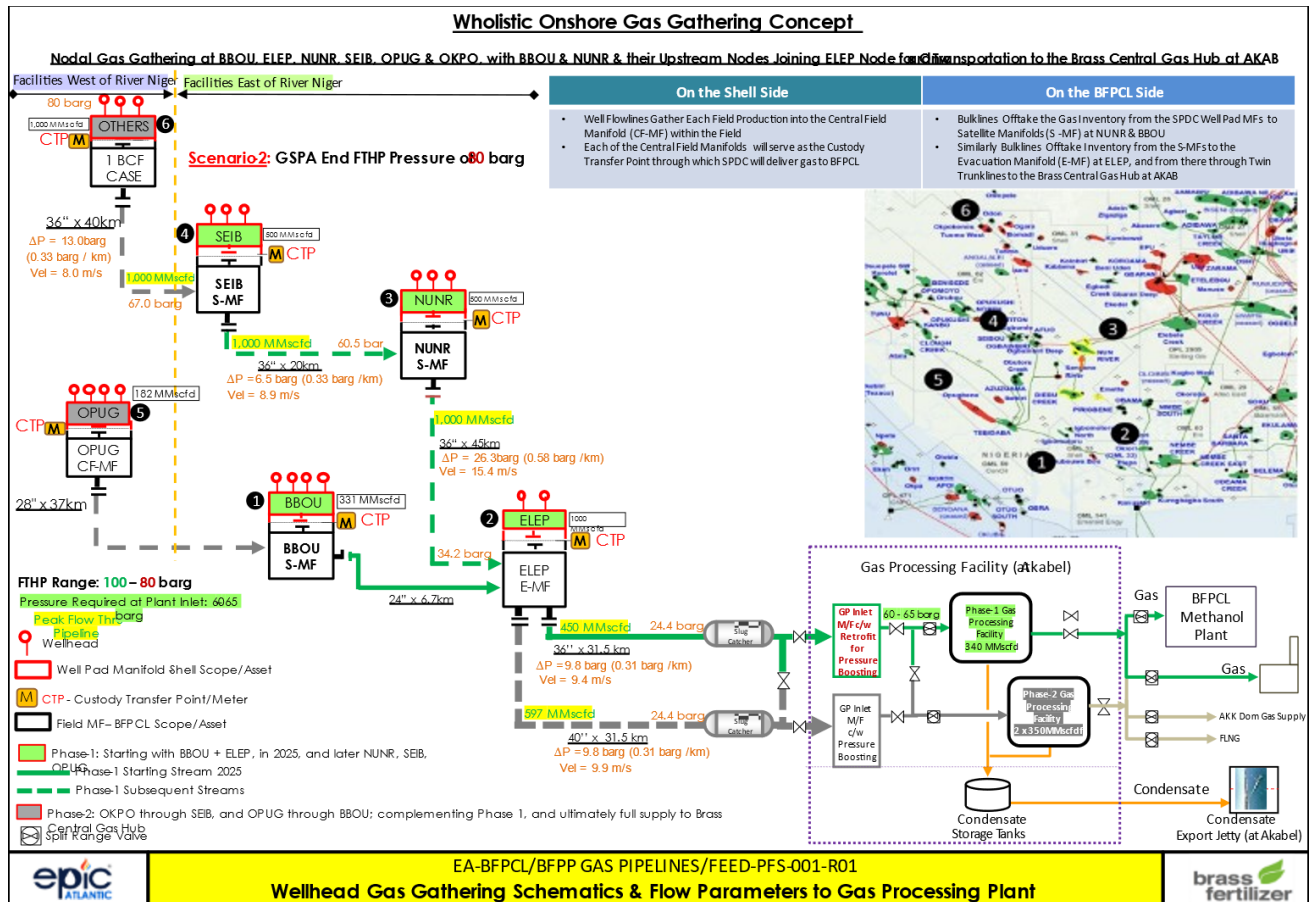


Fig 6.2: Recommended pipeline schematics